

Homework Test 7, Third Question

Let x, y, z be (non-zero) vectors and suppose $w = 2x + 6y + z$.

If $z = -x - 3y$, then $w =$ $x +$ y .

Using the calculation above, mark the statements below that must be true.

- ☒ A. $\text{Span}(x, z) = \text{Span}(y, z)$
- ☒ B. $\text{Span}(x, y, z) = \text{Span}(w, x)$
- ☐ C. $\text{Span}(w, z) = \text{Span}(y, z)$
- ☐ D. $\text{Span}(x, z) = \text{Span}(w, z)$
- ☒ E. $\text{Span}(w, y) = \text{Span}(w, y, z)$

A.) Must be true:

$$\text{span}(x, z) = \text{span}(x, -x - 3y) = \text{span}(x, y)$$

and

$$\text{span}(y, z) = \text{span}(y, -x - 3y) = \text{span}(y, x)$$

equal

B.) Must be true:

$$\text{span}(x, y, z) = \text{span}(x, y, -x - 3y) = \text{span}(x, y)$$

and

$$\text{span}(w, x) = \text{span}(x + 3y, x) = \text{span}(x, y)$$

equal

C.)

$$\text{span}(w, z) = \text{span}(x + 3y, -x - 3y) = \text{span}(x + 3y)$$

and

$$\text{span}(y, z) = \text{span}(y, -x - 3y) = \text{span}(x, y)$$

in general not equal

Hence does not need to be true.

D.)

$$\text{span}(x, z) = \text{span}(x, -x - 3y) = \text{span}(x, y)$$

and

$$\text{span}(w, z) = \text{span}(x + 3y, -x - 3y) = \text{span}(x + 3y)$$

in general not equal

Hence does not need to be true.

E.) Must be true:

$$\text{span}(w, y) = \text{span}(x + 3y, y) = \text{span}(x, y)$$

and

$$\text{span}(w, y, z) = \text{span}(x + 3y, y, -x - 3y) = \text{span}(x + 3y, y) = \text{span}(x, y)$$

equal

Let $\mathbf{u}_4$ be a vector that is not a linear combination of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \}$.

Select the best statement.

Does not hold because u_4 is not a linear combination of $\{u_1, u_2, u_3\}$.

- ☐ A. $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.
- ☐ B. $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a proper subset of $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.
- ☐ C. We only know that $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} \subseteq \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. This does *not* hold.
- ☐ D. There is no obvious relationship between $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ Wrong statement
- ☐ E. $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ when none of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \}$ is a linear combination of the others.
- ☐ F. We only know that $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \subseteq \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$. (B) is a stronger statement. Wrong statement
- ☐ G. none of the above Wrong

B is the correct answer:
 $\text{span}(u_1, u_2, u_3)$ is of course a subset of $\text{span}(u_1, u_2, u_3, u_4)$
And it is a proper subset because (for instance)
 $u_4 \notin \text{span}(\{u_1, u_2, u_3\})$.